

X-ray phase-contrast imaging

“...early-stage tumors and associated vascularization can be visualized via this imaging scheme”

Introduction

As the selection of high-sensitivity scientific detectors, custom phosphor screens, and advanced x-ray sources available to researchers continues to expand, so too does the scope and variety of x-ray phase-contrast imaging techniques. This note describes a new application of phase-contrast imaging in which high-performance Princeton Instruments Quad-RO and PI-SCX cameras are utilized to acquire high-resolution, quantitative x-ray images.

X-ray phase-contrast imaging

X-ray phase-contrast imaging is an important method for visualizing cellular and histological structures for a wide range of biological and medical studies. While traditional x-ray imaging yields an image that maps a sample's absorptive properties by measuring x-ray photon flux from an x-ray source after it traverses a sample, x-ray phase-contrast imaging uses a spatially coherent beam and a high-resolution detector to acquire a clearer, more detailed image of the sample. As the beam's coherent

x-ray photons traverse regions of differing indices of x-ray refraction in a sample, they are refracted and undergo a phase shift, thereby losing coherence and creating constructive and destructive interference patterns with unrefracted photons (see Figure 1). These patterns enable high-contrast imaging of interfaces within the sample.¹

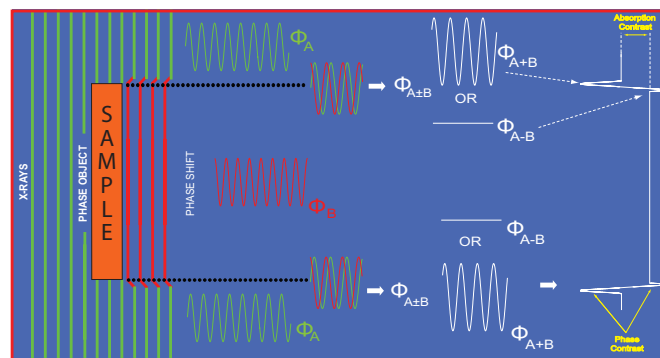


Figure 1. Phase interference effects at edges of the phase object enhance absorption contrast. Diagram courtesy of Dr. Christoph Rose-Petruck, Department of Chemistry, Brown University.

System description

Recently, the groups of Dr. Christoph Rose-Petruck and Dr. Gerald Diebold at Brown University and researchers from the Liver Research Center, Rhode Island Hospital and Warren Alpert Medical School of Brown University, developed an experiment protocol in which a microfocus x-ray source and a scientific detector are

used to produce high-resolution, phase-contrast images of vasculature in fixed mouse livers. Their collaboration with colleagues at the Illinois Institute of Technology subsequently expanded this work to phase-contrast, computed tomographic imaging of fixed murine livers.¹

This Princeton Instruments application note briefly details the Rose-Petruck/Diebold Imaging Group setup and presents some of the data acquired.

X-ray source

A microfocus x-ray source is used to accelerate electrons to 90 keV at a focused point on a tungsten anode 20 μm in diameter. The deceleration of the electrons as they interact with the anode produces polychromatic x-radiation emitted from the 20 μm electron beam focus. The focus size is the limit of the resolution and spatial coherence. Typically, R1 is ~ 60 cm, R2 is ~ 180 cm, and the magnification is $\sim 3\times$ (see Figure 2).

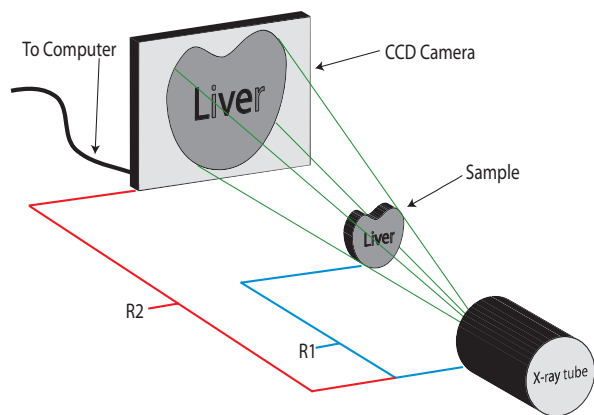


Figure 2. X-ray phase-contrast imaging system setup. Diagram courtesy of Dr. Christoph Rose-Petruck, Department of Chemistry, Brown University.

Detector

To reduce x-ray absorption in air, the x-ray beam traverses a helium environment and impinges on a GdOS:Tb phosphor-coupled CCD. The CCD camera is a thermoelectrically cooled, 16-megapixel Princeton Instruments PI-SCX with 15 μm pixels arranged in a 4096 x 4096 array.

Protocol and results

Excised samples are immersed in formaldehyde, which cross-links the proteins in the liver. Then the samples are dehumidified by less than 20% to allow the vessels and microvessels to fill with air. The density gradient at the air/soft-tissue interface generates strong x-ray absorption and phase-contrast features.

The resultant vascular images are resolved down to the 20 μm scale (see Figure 3), encompassing the size of the smallest vessels. Other images, taken with water-filled vessels, exhibit the same resolution but require longer exposure times.

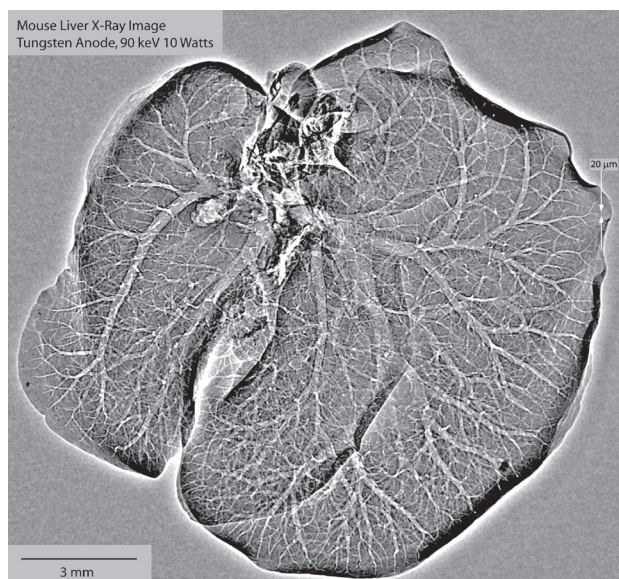


Figure 3. Data acquired with a Princeton Instruments PI-SCX:4096 camera using x-ray phase-contrast imaging. Image courtesy of Dr. Christoph Rose-Petruck, Department of Chemistry, Brown University.

By employing a murine model of human colon cancer metastasized into the liver, early-stage tumors and associated vascularization can be visualized via this imaging scheme.¹ Vascular growth and necrosis play a key role in the proliferation dynamics of several human pathologies, explain the researchers. Many types of carcinomas signal neovascularization from nearby blood vessels, giving the tumor access to nutrients essential for growth and metastasis. Therefore, the researchers note, vascular imaging is of critical importance for studying vascular growth dynamics and potentially determining the efficacy of therapeutic drugs aimed at preventing neovascularization.

Future applications

In addition to the PI-SCX camera referenced in this experiment protocol, the Rose-Petruck/Diebold Imaging Group now utilizes a Princeton Instruments Quad-RO camera. Like the PI-SCX, the Quad-RO features a thermoelectrically cooled CCD to deliver good quantum efficiency (QE) at 550 nm. Also like its PI-SCX predecessor, the Quad-RO can be configured with a high-resolution photosensitive array of 4096 x 4096 pixels, each measuring 15 μm , that affords a full 61 mm x 61 mm field of view.

The Quad-RO, however, offers researchers several new performance advantages. Perhaps most significantly, while the PI-SCX provides single-port readout, the Quad-RO boasts four-port readout. Electronically balanced quadrants, matched within 1.0%, yield

extremely uniform raw images. Dr. Rose-Petruck reports a readout time of ~ 4 sec per full-resolution image with the Quad-RO, as opposed to ~ 17 sec per full-resolution image with the PI-SCX.

He goes on to indicate that a likely next step in his laboratory will be the development of a similar protocol for *ex vivo* perfusion imaging of livers. Another potential application involves the simultaneous use of ultrasound (structural differences in tissue are characterized by acoustic differences) to further enhance the x-ray image contrast. Highly advanced scientific detectors such as the Quad-RO x-ray camera can help facilitate progress in emerging research areas such as these.

Quad-RO

Designed for indirect imaging of x-rays or other Lambertian sources, the Princeton Instruments Quad-RO camera (see Figure 4) not only offers the ability to read out at four ports, it provides two readout speeds per port, on-board memory for loss-free images, and an industry-standard FireWire (IEEE-1394a) interface.



Figure 4. Multiport Princeton Instruments Quad-RO camera.

Software-selectable, dual-speed operation (500 kHz or 1 MHz) and multiple gain settings allow researchers to tailor Quad-RO imaging performance for practically any demanding medium-energy (3.5 keV – 150 keV) x-ray application.

Phosphor screens

The Quad-RO camera's industry-unique mechanical design, in which the fiberoptic extends outside the vacuum, offers the flexibility needed to optimize system performance at the desired x-ray energy *in the lab* with custom phosphor screens.

GdOS:Tb phosphor screens are available for 8, 12, and 17 keV at an emission wavelength of 550 nm. CsI:Tl phosphor screens are also offered.

References

1. "High-resolution angiography: Computed tomography coupled X-ray phase contrast imaging of excised murine liver samples," C. M. Laperle^a, P. Wintermeyer^b, E. Walker^a, D. Shi^c, M. Anastasio^c, C. Rose-Petruck^a, G. Diebold^a, and J. R. Wands^b, *Physics in Medicine and Biology* **53**, 6911-6923 (2008).

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Resources

For more information about high-performance x-ray cameras from Princeton Instruments, please visit:
www.princetoninstruments.com

For more information about Dr. Rose-Petruck's research, please visit: **www.rosepetruck.chem.brown.edu**

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